Newman-Penrose Formalism Calculations

Stephani et al., Equation (36.37)

Metric

$$ds^2 = -A\mathrm{d}t \otimes \mathrm{d}t + B\mathrm{d}r \otimes \mathrm{d}r + C\mathrm{d}\theta \otimes \mathrm{d}\theta + HM^2\mathrm{d}\phi \otimes \mathrm{d}\phi$$

Coordinates

$$(x^\mu) = (t, r, \theta, \phi)$$

Metric Functions

$$A(t, r, \theta, \phi) = A$$

$$B(t, r) = B$$

$$C(t, r) = C$$

$$M(\theta) = M$$

$$H(t, r) = H$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2}\sqrt{2}\sqrt{A}\right) dt + \left(\frac{1}{2}\sqrt{2}\sqrt{B}\right) dr$$

$$n = \left(\frac{1}{2}\sqrt{2}\sqrt{A}\right) dt + \left(-\frac{1}{2}\sqrt{2}\sqrt{B}\right) dr$$

$$m = \left(\frac{1}{2}\sqrt{2}\sqrt{C}\right) d\theta + \left(\frac{1}{2} i \sqrt{2}\sqrt{H} M\right) d\phi$$

$$\overline{m} = \left(\frac{1}{2}\sqrt{2}\sqrt{C}\right) d\theta + \left(-\frac{1}{2} i \sqrt{2}\sqrt{H} M\right) d\phi$$

Contravariant components

$$l = \left(-\frac{\sqrt{2}}{2\sqrt{A}}\right) dt + \left(\frac{\sqrt{2}}{2\sqrt{B}}\right) dr$$

$$n = \left(-\frac{\sqrt{2}}{2\sqrt{A}}\right) dt + \left(-\frac{\sqrt{2}}{2\sqrt{B}}\right) dr$$

$$m = \left(\frac{\sqrt{2}}{2\sqrt{C}}\right) d\theta + \left(\frac{i\sqrt{2}}{2\sqrt{H} M}\right) d\phi$$

$$\overline{m} = \left(\frac{\sqrt{2}}{2\sqrt{C}}\right) d\theta + \left(-\frac{i\sqrt{2}}{2\sqrt{H} M}\right) d\phi$$

Directional Derivatives

$$D = -\frac{\sqrt{2}}{2\sqrt{A}}\,\partial_t + \frac{\sqrt{2}}{2\sqrt{B}}\,\partial_r$$

$$\Delta = -\frac{\sqrt{2}}{2\sqrt{A}}\,\partial_t - \frac{\sqrt{2}}{2\sqrt{B}}\,\partial_r$$

$$\delta = \frac{\sqrt{2}}{2\sqrt{C}}\,\partial_\theta + \frac{i\sqrt{2}}{2\sqrt{H}M}\,\partial_\phi$$

$$\overline{\delta} = \frac{\sqrt{2}}{2\sqrt{C}}\,\partial_\theta - \frac{i\sqrt{2}}{2\sqrt{H}M}\,\partial_\phi$$

Spin Coefficients

$$\rho = \frac{\sqrt{2}\frac{\partial}{\partial t}C}{8\sqrt{A}C} - \frac{\sqrt{2}\frac{\partial}{\partial r}C}{8\sqrt{B}C} + \frac{\sqrt{2}\frac{\partial}{\partial t}H}{8\sqrt{A}H} - \frac{\sqrt{2}\frac{\partial}{\partial r}H}{8\sqrt{B}H}$$

$$\sigma = \frac{\sqrt{2}\frac{\partial}{\partial t}C}{8\sqrt{A}C} - \frac{\sqrt{2}\frac{\partial}{\partial r}C}{8\sqrt{B}C} - \frac{\sqrt{2}\frac{\partial}{\partial t}H}{8\sqrt{A}H} + \frac{\sqrt{2}\frac{\partial}{\partial r}H}{8\sqrt{B}H}$$

$$\kappa = -\frac{\sqrt{2}\frac{\partial}{\partial \theta}A}{8\,A\sqrt{C}} - \frac{i\sqrt{2}\frac{\partial}{\partial \phi}A}{8\,A\sqrt{H}M}$$

$$\mu = -\frac{\sqrt{2}\frac{\partial}{\partial t}C}{8\sqrt{A}C} - \frac{\sqrt{2}\frac{\partial}{\partial r}C}{8\sqrt{B}C} - \frac{\sqrt{2}\frac{\partial}{\partial t}H}{8\sqrt{A}H} - \frac{\sqrt{2}\frac{\partial}{\partial r}H}{8\sqrt{B}H}$$

$$\lambda = -\frac{\sqrt{2}\frac{\partial}{\partial t}C}{8\sqrt{A}C} - \frac{\sqrt{2}\frac{\partial}{\partial r}C}{8\sqrt{B}C} + \frac{\sqrt{2}\frac{\partial}{\partial t}H}{8\sqrt{A}H} + \frac{\sqrt{2}\frac{\partial}{\partial r}H}{8\sqrt{B}H}$$

$$\nu = \frac{\sqrt{2}\frac{\partial}{\partial \theta}A}{8\,A\sqrt{C}} - \frac{i\sqrt{2}\frac{\partial}{\partial \phi}A}{8\,A\sqrt{H}M}$$

$$\alpha = -\frac{\sqrt{2}\frac{\partial}{\partial \theta}M}{4\sqrt{C}M}$$

$$\beta = \frac{\sqrt{2}\frac{\partial}{\partial\theta}M}{4\sqrt{C}M}$$

$$\tau = -\frac{\sqrt{2}\frac{\partial}{\partial\theta}A}{8A\sqrt{C}} - \frac{i\sqrt{2}\frac{\partial}{\partial\phi}A}{8A\sqrt{H}M}$$

$$\pi = \frac{\sqrt{2}\frac{\partial}{\partial\theta}A}{8A\sqrt{C}} - \frac{i\sqrt{2}\frac{\partial}{\partial\phi}A}{8A\sqrt{H}M}$$

$$\epsilon = \frac{\sqrt{2}\frac{\partial}{\partial r}A}{8A\sqrt{B}} - \frac{\sqrt{2}\frac{\partial}{\partial t}B}{8\sqrt{A}B}$$

$$\gamma = \frac{\sqrt{2}\frac{\partial}{\partial r}A}{8A\sqrt{B}} + \frac{\sqrt{2}\frac{\partial}{\partial t}B}{8\sqrt{A}B}$$

Ricci Tensor Components

$$\begin{aligned}\Phi_{00} = &-\frac{\frac{\partial}{\partial\theta}A^2}{16A^2C} + \frac{\frac{\partial^2}{(\partial\theta)^2}A}{8AC} + \frac{\frac{\partial}{\partial t}A\frac{\partial}{\partial t}C}{16A^2C} - \frac{\frac{\partial}{\partial r}A\frac{\partial}{\partial t}C}{8A^{\frac{3}{2}}\sqrt{B}C} + \frac{\frac{\partial}{\partial t}B\frac{\partial}{\partial t}C}{16ABC} + \frac{\frac{\partial}{\partial t}C^2}{16AC^2} - \frac{\frac{\partial^2}{(\partial t)^2}C}{8AC} + \frac{\frac{\partial^2}{\partial t\partial r}C}{4\sqrt{A}\sqrt{B}C} + \frac{\frac{\partial}{\partial r}A\frac{\partial}{\partial r}C}{16ABC} - \frac{\frac{\partial}{\partial t}B\frac{\partial}{\partial r}C}{8\sqrt{A}B^{\frac{3}{2}}C} + \frac{\frac{\partial}{\partial r}B\frac{\partial}{\partial r}C}{16B^2C} - \frac{\frac{\partial}{\partial t}C\frac{\partial}{\partial r}C}{8\sqrt{A}\sqrt{B}C^2} + \frac{\frac{\partial}{\partial r}C^2}{16BC^2} - \frac{\frac{\partial^2}{(\partial r)^2}C}{8BC} + \frac{\frac{\partial}{\partial t}A\frac{\partial}{\partial t}H}{16A^2H} \\ &-\frac{\frac{\partial}{\partial r}A\frac{\partial}{\partial t}H}{8A^{\frac{3}{2}}\sqrt{B}H} + \frac{\frac{\partial}{\partial t}B\frac{\partial}{\partial t}H}{16ABH} + \frac{\frac{\partial}{\partial t}H^2}{16AH^2} - \frac{\frac{\partial^2}{(\partial t)^2}H}{8AH} + \frac{\frac{\partial^2}{\partial t\partial r}H}{4\sqrt{A}\sqrt{B}H} + \frac{\frac{\partial}{\partial r}A\frac{\partial}{\partial r}H}{16ABH} - \frac{\frac{\partial}{\partial t}B\frac{\partial}{\partial r}H}{8\sqrt{A}B^{\frac{3}{2}}H} + \frac{\frac{\partial}{\partial r}B\frac{\partial}{\partial r}H}{16B^2H} - \frac{\frac{\partial}{\partial t}H\frac{\partial}{\partial r}H}{8\sqrt{A}\sqrt{B}H^2} + \frac{\frac{\partial}{\partial r}H^2}{16BH^2} - \frac{\frac{\partial^2}{(\partial r)^2}H}{8BH} + \frac{\frac{\partial}{\partial\theta}A\frac{\partial}{\partial\theta}M}{8ACM} - \frac{\frac{\partial}{\partial\phi}A^2}{16A^2HM^2} + \frac{\frac{\partial^2}{(\partial\phi)^2}A}{8AHM^2}\end{aligned}$$

$$\begin{aligned}\Phi_{01} = &-\frac{\frac{\partial^2}{\partial r\partial\theta}A}{8A\sqrt{B}\sqrt{C}} + \frac{\frac{\partial}{\partial r}A\frac{\partial}{\partial\theta}A}{16A^2\sqrt{B}\sqrt{C}} - \frac{\frac{\partial}{\partial\theta}A\frac{\partial}{\partial t}B}{16A^{\frac{3}{2}}B\sqrt{C}} + \frac{\frac{\partial}{\partial\theta}A\frac{\partial}{\partial r}C}{16A\sqrt{B}C^{\frac{3}{2}}} - \frac{\frac{\partial}{\partial\theta}A\frac{\partial}{\partial t}H}{16A^{\frac{3}{2}}\sqrt{C}H} - \frac{\frac{\partial}{\partial t}C\frac{\partial}{\partial\theta}M}{8\sqrt{A}C^{\frac{3}{2}}M} + \frac{\frac{\partial}{\partial r}C\frac{\partial}{\partial\theta}M}{8\sqrt{B}C^{\frac{3}{2}}M} + \frac{\frac{\partial}{\partial t}H\frac{\partial}{\partial\theta}M}{8\sqrt{A}\sqrt{C}HM} \\ &-\frac{\frac{\partial}{\partial r}H\frac{\partial}{\partial\theta}M}{8\sqrt{B}\sqrt{C}HM} - \frac{i\frac{\partial^2}{\partial r\partial\phi}A}{8A\sqrt{B}\sqrt{H}M} + \frac{i\frac{\partial}{\partial r}A\frac{\partial}{\partial\phi}A}{16A^2\sqrt{B}\sqrt{H}M} - \frac{i\frac{\partial}{\partial\phi}A\frac{\partial}{\partial t}B}{16A^{\frac{3}{2}}B\sqrt{H}M} - \frac{i\frac{\partial}{\partial\phi}A\frac{\partial}{\partial t}C}{16A^{\frac{3}{2}}C\sqrt{H}M} + \frac{i\frac{\partial}{\partial\phi}A\frac{\partial}{\partial r}H}{16A\sqrt{B}H^{\frac{3}{2}}M}\end{aligned}$$

$$\begin{aligned}\Phi_{02} = &\frac{\frac{\partial}{\partial\theta}A^2}{16A^2C} - \frac{\frac{\partial^2}{(\partial\theta)^2}A}{8AC} - \frac{\frac{\partial}{\partial t}A\frac{\partial}{\partial t}C}{16A^2C} + \frac{\frac{\partial}{\partial t}B\frac{\partial}{\partial t}C}{16ABC} - \frac{\frac{\partial}{\partial t}C^2}{16AC^2} + \frac{\frac{\partial^2}{(\partial t)^2}C}{8AC} - \frac{\frac{\partial}{\partial r}A\frac{\partial}{\partial r}C}{16ABC} + \frac{\frac{\partial}{\partial r}B\frac{\partial}{\partial r}C}{16B^2C} + \frac{\frac{\partial}{\partial r}C^2}{16BC^2} - \frac{\frac{\partial^2}{(\partial r)^2}C}{8BC} + \frac{\frac{\partial}{\partial t}A\frac{\partial}{\partial t}H}{16A^2H} - \frac{\frac{\partial}{\partial t}B\frac{\partial}{\partial t}H}{16ABH} + \frac{\frac{\partial}{\partial t}H^2}{16AH^2} \\ &-\frac{\frac{\partial^2}{(\partial t)^2}H}{8AH} + \frac{\frac{\partial}{\partial r}A\frac{\partial}{\partial r}H}{16ABH} - \frac{\frac{\partial}{\partial r}B\frac{\partial}{\partial r}H}{16B^2H} - \frac{\frac{\partial}{\partial r}H^2}{16BH^2} + \frac{\frac{\partial^2}{(\partial r)^2}H}{8BH} + \frac{\frac{\partial}{\partial\theta}A\frac{\partial}{\partial\theta}M}{8ACM} - \frac{i\frac{\partial^2}{\partial\theta\partial\phi}A}{4A\sqrt{C}\sqrt{H}M} + \frac{i\frac{\partial}{\partial\theta}A\frac{\partial}{\partial\phi}A}{8A^2\sqrt{C}\sqrt{H}M} + \frac{i\frac{\partial}{\partial\phi}A\frac{\partial}{\partial\theta}M}{4A\sqrt{C}\sqrt{H}M^2} - \frac{\frac{\partial}{\partial\phi}A^2}{16A^2HM^2} + \frac{\frac{\partial^2}{(\partial\phi)^2}A}{8AHM^2}\end{aligned}$$

$$\Phi_{10} = -\frac{\frac{\partial^2}{\partial r\partial\theta}A}{8\,A\sqrt{B}\sqrt{C}} + \frac{\frac{\partial}{\partial r}A\frac{\partial}{\partial\theta}A}{16\,A^2\sqrt{B}\sqrt{C}} - \frac{\frac{\partial}{\partial\theta}A\frac{\partial}{\partial t}B}{16\,A^{\frac{3}{2}}B\sqrt{C}} + \frac{\frac{\partial}{\partial\theta}A\frac{\partial}{\partial r}C}{16\,A\sqrt{B}C^{\frac{3}{2}}} - \frac{\frac{\partial}{\partial\theta}A\frac{\partial}{\partial t}H}{16\,A^{\frac{3}{2}}\sqrt{C}H} - \frac{\frac{\partial}{\partial t}C\frac{\partial}{\partial\theta}M}{8\,\sqrt{A}C^{\frac{3}{2}}M} + \frac{\frac{\partial}{\partial r}C\frac{\partial}{\partial\theta}M}{8\,\sqrt{B}C^{\frac{3}{2}}M} + \frac{\frac{\partial}{\partial t}H\frac{\partial}{\partial\theta}M}{8\,\sqrt{A}\sqrt{C}HM}$$
$$- \frac{\frac{\partial}{\partial r}H\frac{\partial}{\partial\theta}M}{8\,\sqrt{B}\sqrt{C}HM} + \frac{i\,\frac{\partial^2}{\partial r\partial\phi}A}{8\,A\sqrt{B}\sqrt{H}M} - \frac{i\,\frac{\partial}{\partial r}A\frac{\partial}{\partial\phi}A}{16\,A^2\sqrt{B}\sqrt{H}M} + \frac{i\,\frac{\partial}{\partial\phi}A\frac{\partial}{\partial t}B}{16\,A^{\frac{3}{2}}B\sqrt{H}M} + \frac{i\,\frac{\partial}{\partial\phi}A\frac{\partial}{\partial t}C}{16\,A^{\frac{3}{2}}C\sqrt{H}M} - \frac{i\,\frac{\partial}{\partial\phi}A\frac{\partial}{\partial r}H}{16\,A\sqrt{B}H^{\frac{3}{2}}M}$$

$$\Phi_{11} = -\frac{\frac{\partial}{\partial r}A^2}{16\,A^2B} + \frac{\frac{\partial^2}{(\partial r)^2}A}{8\,AB} + \frac{\frac{\partial}{\partial t}A\frac{\partial}{\partial t}B}{16\,A^2B} + \frac{\frac{\partial}{\partial t}B^2}{16\,AB^2} - \frac{\frac{\partial^2}{(\partial t)^2}B}{8\,AB} - \frac{\frac{\partial}{\partial r}A\frac{\partial}{\partial r}B}{16\,AB^2} + \frac{\frac{\partial}{\partial t}C\frac{\partial}{\partial t}H}{16\,ACH} - \frac{\frac{\partial}{\partial r}C\frac{\partial}{\partial r}H}{16\,BCH} - \frac{\frac{\partial^2}{(\partial\theta)^2}M}{4\,CM}$$

$$\Phi_{12} = \frac{\frac{\partial^2}{\partial r\partial\theta}A}{8\,A\sqrt{B}\sqrt{C}} - \frac{\frac{\partial}{\partial r}A\frac{\partial}{\partial\theta}A}{16\,A^2\sqrt{B}\sqrt{C}} - \frac{\frac{\partial}{\partial\theta}A\frac{\partial}{\partial t}B}{16\,A^{\frac{3}{2}}B\sqrt{C}} - \frac{\frac{\partial}{\partial\theta}A\frac{\partial}{\partial r}C}{16\,A\sqrt{B}C^{\frac{3}{2}}} - \frac{\frac{\partial}{\partial\theta}A\frac{\partial}{\partial t}H}{16\,A^{\frac{3}{2}}\sqrt{C}H} - \frac{\frac{\partial}{\partial t}C\frac{\partial}{\partial\theta}M}{8\,\sqrt{A}C^{\frac{3}{2}}M} - \frac{\frac{\partial}{\partial r}C\frac{\partial}{\partial\theta}M}{8\,\sqrt{B}C^{\frac{3}{2}}M} + \frac{\frac{\partial}{\partial t}H\frac{\partial}{\partial\theta}M}{8\,\sqrt{A}\sqrt{C}HM}$$
$$+ \frac{\frac{\partial}{\partial r}H\frac{\partial}{\partial\theta}M}{8\,\sqrt{B}\sqrt{C}HM} + \frac{i\,\frac{\partial^2}{\partial r\partial\phi}A}{8\,A\sqrt{B}\sqrt{H}M} - \frac{i\,\frac{\partial}{\partial r}A\frac{\partial}{\partial\phi}A}{16\,A^2\sqrt{B}\sqrt{H}M} - \frac{i\,\frac{\partial}{\partial\phi}A\frac{\partial}{\partial t}B}{16\,A^{\frac{3}{2}}B\sqrt{H}M} - \frac{i\,\frac{\partial}{\partial\phi}A\frac{\partial}{\partial t}C}{16\,A^{\frac{3}{2}}C\sqrt{H}M} - \frac{i\,\frac{\partial}{\partial\phi}A\frac{\partial}{\partial r}H}{16\,A\sqrt{B}H^{\frac{3}{2}}M}$$

$$\Phi_{20} = \frac{\frac{\partial}{\partial\theta}A^2}{16\,A^2C} - \frac{\frac{\partial^2}{(\partial\theta)^2}A}{8\,AC} - \frac{\frac{\partial}{\partial t}A\frac{\partial}{\partial t}C}{16\,A^2C} + \frac{\frac{\partial}{\partial t}B\frac{\partial}{\partial t}C}{16\,ABC} - \frac{\frac{\partial}{\partial t}C^2}{16\,AC^2} + \frac{\frac{\partial^2}{(\partial t)^2}C}{8\,AC} - \frac{\frac{\partial}{\partial r}A\frac{\partial}{\partial r}C}{16\,ABC} + \frac{\frac{\partial}{\partial r}B\frac{\partial}{\partial r}C}{16\,B^2C} + \frac{\frac{\partial}{\partial r}C^2}{16\,BC^2} - \frac{\frac{\partial^2}{(\partial r)^2}C}{8\,BC} + \frac{\frac{\partial}{\partial t}A\frac{\partial}{\partial t}H}{16\,A^2H} - \frac{\frac{\partial}{\partial t}B\frac{\partial}{\partial t}H}{16\,ABH} + \frac{\frac{\partial}{\partial t}H^2}{16\,AH^2}$$
$$- \frac{\frac{\partial^2}{(\partial t)^2}H}{8\,AH} + \frac{\frac{\partial}{\partial r}A\frac{\partial}{\partial r}H}{16\,ABH} - \frac{\frac{\partial}{\partial r}B\frac{\partial}{\partial r}H}{16\,B^2H} - \frac{\frac{\partial}{\partial r}H^2}{16\,BH^2} + \frac{\frac{\partial^2}{(\partial r)^2}H}{8\,BH} + \frac{\frac{\partial}{\partial\theta}A\frac{\partial}{\partial\theta}M}{8\,ACM} + \frac{i\,\frac{\partial^2}{\partial\theta\partial\phi}A}{4\,A\sqrt{C}\sqrt{H}M} - \frac{i\,\frac{\partial}{\partial\theta}A\frac{\partial}{\partial\phi}A}{8\,A^2\sqrt{C}\sqrt{H}M} - \frac{i\,\frac{\partial}{\partial\phi}A\frac{\partial}{\partial\theta}M}{4\,A\sqrt{C}\sqrt{H}M^2} - \frac{\frac{\partial}{\partial\phi}A^2}{16\,A^2HM^2} + \frac{\frac{\partial^2}{(\partial\phi)^2}A}{8\,AHM^2}$$

$$\Phi_{21} = \frac{\frac{\partial^2}{\partial r\partial\theta}A}{8\,A\sqrt{B}\sqrt{C}} - \frac{\frac{\partial}{\partial r}A\frac{\partial}{\partial\theta}A}{16\,A^2\sqrt{B}\sqrt{C}} - \frac{\frac{\partial}{\partial\theta}A\frac{\partial}{\partial t}B}{16\,A^{\frac{3}{2}}B\sqrt{C}} - \frac{\frac{\partial}{\partial\theta}A\frac{\partial}{\partial r}C}{16\,A\sqrt{B}C^{\frac{3}{2}}} - \frac{\frac{\partial}{\partial\theta}A\frac{\partial}{\partial t}H}{16\,A^{\frac{3}{2}}\sqrt{C}H} - \frac{\frac{\partial}{\partial t}C\frac{\partial}{\partial\theta}M}{8\,\sqrt{A}C^{\frac{3}{2}}M} - \frac{\frac{\partial}{\partial r}C\frac{\partial}{\partial\theta}M}{8\,\sqrt{B}C^{\frac{3}{2}}M} + \frac{\frac{\partial}{\partial t}H\frac{\partial}{\partial\theta}M}{8\,\sqrt{A}\sqrt{C}HM}$$
$$+ \frac{\frac{\partial}{\partial r}H\frac{\partial}{\partial\theta}M}{8\,\sqrt{B}\sqrt{C}HM} - \frac{i\,\frac{\partial^2}{\partial r\partial\phi}A}{8\,A\sqrt{B}\sqrt{H}M} + \frac{i\,\frac{\partial}{\partial r}A\frac{\partial}{\partial\phi}A}{16\,A^2\sqrt{B}\sqrt{H}M} + \frac{i\,\frac{\partial}{\partial\phi}A\frac{\partial}{\partial t}B}{16\,A^{\frac{3}{2}}B\sqrt{H}M} + \frac{i\,\frac{\partial}{\partial\phi}A\frac{\partial}{\partial t}C}{16\,A^{\frac{3}{2}}C\sqrt{H}M} + \frac{i\,\frac{\partial}{\partial\phi}A\frac{\partial}{\partial r}H}{16\,A\sqrt{B}H^{\frac{3}{2}}M}$$

$$\Phi_{22} = -\frac{\frac{\partial}{\partial\theta}A^2}{16\,A^2C} + \frac{\frac{\partial^2}{(\partial\theta)^2}A}{8\,AC} + \frac{\frac{\partial}{\partial t}A\frac{\partial}{\partial t}C}{16\,A^2C} + \frac{\frac{\partial}{\partial r}A\frac{\partial}{\partial t}C}{8\,A^{\frac{3}{2}}\sqrt{B}C} + \frac{\frac{\partial}{\partial t}B\frac{\partial}{\partial t}C}{16\,ABC} + \frac{\frac{\partial}{\partial t}C^2}{16\,AC^2} - \frac{\frac{\partial^2}{(\partial t)^2}C}{8\,AC} - \frac{\frac{\partial^2}{\partial t\partial r}C}{4\,\sqrt{A}\sqrt{B}C} + \frac{\frac{\partial}{\partial r}A\frac{\partial}{\partial r}C}{16\,ABC} + \frac{\frac{\partial}{\partial t}B\frac{\partial}{\partial r}C}{8\,\sqrt{A}B^{\frac{3}{2}}C} + \frac{\frac{\partial}{\partial r}B\frac{\partial}{\partial r}C}{16\,B^2C} + \frac{\frac{\partial}{\partial t}C\frac{\partial}{\partial r}C}{8\,\sqrt{A}\sqrt{B}C^2} + \frac{\frac{\partial}{\partial r}C^2}{16\,BC^2} - \frac{\frac{\partial^2}{(\partial r)^2}C}{8\,BC} + \frac{\frac{\partial}{\partial t}A\frac{\partial}{\partial t}H}{16\,A^2H}$$
$$+ \frac{\frac{\partial}{\partial r}A\frac{\partial}{\partial t}H}{8\,A^{\frac{3}{2}}\sqrt{B}H} + \frac{\frac{\partial}{\partial t}B\frac{\partial}{\partial t}H}{16\,ABH} + \frac{\frac{\partial}{\partial t}H^2}{16\,AH^2} - \frac{\frac{\partial^2}{(\partial t)^2}H}{8\,AH} - \frac{\frac{\partial^2}{\partial t\partial r}H}{4\,\sqrt{A}\sqrt{B}H} + \frac{\frac{\partial}{\partial r}A\frac{\partial}{\partial r}H}{16\,ABH} + \frac{\frac{\partial}{\partial t}B\frac{\partial}{\partial r}H}{8\,\sqrt{A}B^{\frac{3}{2}}H} + \frac{\frac{\partial}{\partial r}B\frac{\partial}{\partial r}H}{16\,B^2H} + \frac{\frac{\partial}{\partial t}H\frac{\partial}{\partial r}H}{8\,\sqrt{A}\sqrt{B}H^2} + \frac{\frac{\partial}{\partial r}H^2}{16\,BH^2} - \frac{\frac{\partial^2}{(\partial r)^2}H}{8\,BH} + \frac{\frac{\partial}{\partial\theta}A\frac{\partial}{\partial\theta}M}{8\,ACM} - \frac{\frac{\partial}{\partial\phi}A^2}{16\,A^2HM^2} + \frac{\frac{\partial^2}{(\partial\phi)^2}A}{8\,AHM^2}$$

$$\Lambda = \frac{\frac{\partial}{\partial r}A^2}{48\,A^2B} - \frac{\frac{\partial^2}{(\partial r)^2}A}{24\,AB} + \frac{\frac{\partial}{\partial\theta}A^2}{48\,A^2C} - \frac{\frac{\partial^2}{(\partial\theta)^2}A}{24\,AC} - \frac{\frac{\partial}{\partial t}A\frac{\partial}{\partial t}B}{48\,A^2B} - \frac{\frac{\partial}{\partial t}B^2}{48\,AB^2} + \frac{\frac{\partial^2}{(\partial t)^2}B}{24\,AB} + \frac{\frac{\partial}{\partial r}A\frac{\partial}{\partial r}B}{48\,AB^2} - \frac{\frac{\partial}{\partial t}A\frac{\partial}{\partial t}C}{48\,A^2C} + \frac{\frac{\partial}{\partial t}B\frac{\partial}{\partial t}C}{48\,ABC} - \frac{\frac{\partial}{\partial t}C^2}{48\,AC^2} + \frac{\frac{\partial^2}{(\partial t)^2}C}{24\,AC} - \frac{\frac{\partial}{\partial r}A\frac{\partial}{\partial r}C}{48\,ABC} + \frac{\frac{\partial}{\partial r}B\frac{\partial}{\partial r}C}{48\,B^2C} + \frac{\frac{\partial}{\partial r}C^2}{48\,BC^2} - \frac{\frac{\partial^2}{(\partial r)^2}C}{24\,BC}$$
$$- \frac{\frac{\partial}{\partial t}A\frac{\partial}{\partial t}H}{48\,A^2H} + \frac{\frac{\partial}{\partial t}B\frac{\partial}{\partial t}H}{48\,ABH} + \frac{\frac{\partial}{\partial t}C\frac{\partial}{\partial t}H}{48\,ACH} - \frac{\frac{\partial}{\partial t}H^2}{48\,AH^2} + \frac{\frac{\partial^2}{(\partial t)^2}H}{24\,AH} - \frac{\frac{\partial}{\partial r}A\frac{\partial}{\partial r}H}{48\,ABH} + \frac{\frac{\partial}{\partial r}B\frac{\partial}{\partial r}H}{48\,B^2H} - \frac{\frac{\partial}{\partial r}C\frac{\partial}{\partial r}H}{48\,BCH} + \frac{\frac{\partial}{\partial r}H^2}{48\,BH^2} - \frac{\frac{\partial^2}{(\partial r)^2}H}{24\,BH} - \frac{\frac{\partial}{\partial\theta}A\frac{\partial}{\partial\theta}M}{24\,ACM} - \frac{\frac{\partial^2}{(\partial\theta)^2}M}{12\,CM} + \frac{\frac{\partial}{\partial\phi}A^2}{48\,A^2HM^2} - \frac{\frac{\partial^2}{(\partial\phi)^2}A}{24\,AHM^2}$$

Weyl Tensor Components

$$\Psi_0 = -\frac{\frac{\partial}{\partial\theta}A^2}{16\,A^2C} + \frac{\frac{\partial^2}{(\partial\theta)^2}A}{8\,AC} + \frac{\frac{\partial}{\partial t}A\frac{\partial}{\partial t}C}{16\,A^2C} - \frac{\frac{\partial}{\partial r}A\frac{\partial}{\partial t}C}{8\,A^{\frac{3}{2}}\sqrt{B}C} + \frac{\frac{\partial}{\partial t}B\frac{\partial}{\partial t}C}{16\,ABC} + \frac{\frac{\partial}{\partial t}C^2}{16\,AC^2} - \frac{\frac{\partial^2}{(\partial t)^2}C}{8\,AC} + \frac{\frac{\partial^2}{\partial t\partial r}C}{4\sqrt{A}\sqrt{B}C} + \frac{\frac{\partial}{\partial r}A\frac{\partial}{\partial r}C}{16\,ABC} - \frac{\frac{\partial}{\partial t}B\frac{\partial}{\partial r}C}{8\sqrt{A}B^{\frac{3}{2}}C} + \frac{\frac{\partial}{\partial r}B\frac{\partial}{\partial r}C}{16\,B^2C}$$
$$- \frac{\frac{\partial}{\partial t}C\frac{\partial}{\partial r}C}{8\sqrt{A}\sqrt{B}C^2} + \frac{\frac{\partial}{\partial r}C^2}{16\,BC^2} - \frac{\frac{\partial^2}{(\partial r)^2}C}{8\,BC} - \frac{\frac{\partial}{\partial t}A\frac{\partial}{\partial t}H}{16\,A^2H} + \frac{\frac{\partial}{\partial r}A\frac{\partial}{\partial t}H}{8\,A^{\frac{3}{2}}\sqrt{B}H} - \frac{\frac{\partial}{\partial t}B\frac{\partial}{\partial t}H}{16\,ABH} - \frac{\frac{\partial}{\partial t}H^2}{16\,AH^2} + \frac{\frac{\partial^2}{(\partial t)^2}H}{8\,AH} - \frac{\frac{\partial^2}{\partial t\partial r}H}{4\sqrt{A}\sqrt{B}H} - \frac{\frac{\partial}{\partial r}A\frac{\partial}{\partial r}H}{16\,ABH} + \frac{\frac{\partial}{\partial t}B\frac{\partial}{\partial r}H}{8\sqrt{A}B^{\frac{3}{2}}H}$$
$$- \frac{\frac{\partial}{\partial r}B\frac{\partial}{\partial r}H}{16\,B^2H} + \frac{\frac{\partial}{\partial t}H\frac{\partial}{\partial r}H}{8\sqrt{A}\sqrt{B}H^2} - \frac{\frac{\partial}{\partial r}H^2}{16\,BH^2} + \frac{\frac{\partial^2}{(\partial r)^2}H}{8\,BH} - \frac{\frac{\partial}{\partial\theta}A\frac{\partial}{\partial\theta}M}{8\,ACM} + \frac{i\,\frac{\partial^2}{\partial\theta\partial\phi}A}{4\,A\sqrt{C}\sqrt{H}M} - \frac{i\,\frac{\partial}{\partial\theta}A\frac{\partial}{\partial\phi}A}{8\,A^2\sqrt{C}\sqrt{H}M} - \frac{i\,\frac{\partial}{\partial\phi}A\frac{\partial}{\partial\theta}M}{4\,A\sqrt{C}\sqrt{H}M^2} + \frac{\frac{\partial}{\partial\phi}A^2}{16\,A^2HM^2} - \frac{\frac{\partial^2}{(\partial\phi)^2}A}{8\,AHM^2}$$

$$\Psi_1 = -\frac{\frac{\partial^2}{\partial r\partial\theta}A}{8\,A\sqrt{B}\sqrt{C}} + \frac{\frac{\partial}{\partial r}A\frac{\partial}{\partial\theta}A}{16\,A^2\sqrt{B}\sqrt{C}} - \frac{\frac{\partial}{\partial\theta}A\frac{\partial}{\partial t}B}{16\,A^{\frac{3}{2}}B\sqrt{C}} + \frac{\frac{\partial}{\partial\theta}A\frac{\partial}{\partial r}C}{16\,A\sqrt{B}C^{\frac{3}{2}}} + \frac{\frac{\partial}{\partial\theta}A\frac{\partial}{\partial t}H}{16\,A^{\frac{3}{2}}\sqrt{C}H} + \frac{\frac{\partial}{\partial t}C\frac{\partial}{\partial\theta}M}{8\sqrt{A}C^{\frac{3}{2}}M} - \frac{\frac{\partial}{\partial r}C\frac{\partial}{\partial\theta}M}{8\sqrt{B}C^{\frac{3}{2}}M} - \frac{\frac{\partial}{\partial t}H\frac{\partial}{\partial\theta}M}{8\sqrt{A}\sqrt{C}HM}$$
$$+ \frac{\frac{\partial}{\partial r}H\frac{\partial}{\partial\theta}M}{8\sqrt{B}\sqrt{C}HM} - \frac{i\,\frac{\partial^2}{\partial r\partial\phi}A}{8\,A\sqrt{B}\sqrt{H}M} + \frac{i\,\frac{\partial}{\partial r}A\frac{\partial}{\partial\phi}A}{16\,A^2\sqrt{B}\sqrt{H}M} - \frac{i\,\frac{\partial}{\partial\phi}A\frac{\partial}{\partial t}B}{16\,A^{\frac{3}{2}}B\sqrt{H}M} + \frac{i\,\frac{\partial}{\partial\phi}A\frac{\partial}{\partial t}C}{16\,A^{\frac{3}{2}}C\sqrt{H}M} + \frac{i\,\frac{\partial}{\partial\phi}A\frac{\partial}{\partial r}H}{16\,A\sqrt{B}H^{\frac{3}{2}}M}$$

$$\Psi_2 = -\frac{\frac{\partial}{\partial r}A^2}{24\,A^2B} + \frac{\frac{\partial^2}{(\partial r)^2}A}{12\,AB} + \frac{\frac{\partial}{\partial\theta}A^2}{48\,A^2C} - \frac{\frac{\partial^2}{(\partial\theta)^2}A}{24\,AC} + \frac{\frac{\partial}{\partial t}A\frac{\partial}{\partial t}B}{24\,A^2B} + \frac{\frac{\partial}{\partial t}B^2}{24\,AB^2} - \frac{\frac{\partial^2}{(\partial t)^2}B}{12\,AB} - \frac{\frac{\partial}{\partial r}A\frac{\partial}{\partial r}B}{24\,AB^2} - \frac{\frac{\partial}{\partial t}A\frac{\partial}{\partial t}C}{48\,A^2C} + \frac{\frac{\partial}{\partial t}B\frac{\partial}{\partial t}C}{48\,ABC} - \frac{\frac{\partial}{\partial t}C^2}{48\,AC^2} + \frac{\frac{\partial^2}{(\partial t)^2}C}{24\,AC} - \frac{\frac{\partial}{\partial r}A\frac{\partial}{\partial r}C}{48\,ABC} + \frac{\frac{\partial}{\partial r}B\frac{\partial}{\partial r}C}{48\,B^2C} + \frac{\frac{\partial}{\partial r}C^2}{48\,BC^2} - \frac{\frac{\partial^2}{(\partial r)^2}C}{24\,BC}$$
$$- \frac{\frac{\partial}{\partial t}A\frac{\partial}{\partial t}H}{48\,A^2H} + \frac{\frac{\partial}{\partial t}B\frac{\partial}{\partial t}H}{48\,ABH} - \frac{\frac{\partial}{\partial t}C\frac{\partial}{\partial t}H}{24\,ACH} - \frac{\frac{\partial}{\partial t}H^2}{48\,AH^2} + \frac{\frac{\partial^2}{(\partial t)^2}H}{24\,AH} - \frac{\frac{\partial}{\partial r}A\frac{\partial}{\partial r}H}{48\,ABH} + \frac{\frac{\partial}{\partial r}B\frac{\partial}{\partial r}H}{48\,B^2H} + \frac{\frac{\partial}{\partial r}C\frac{\partial}{\partial r}H}{24\,BCH} + \frac{\frac{\partial}{\partial r}H^2}{48\,BH^2} - \frac{\frac{\partial^2}{(\partial r)^2}H}{24\,BH} - \frac{\frac{\partial}{\partial\theta}A\frac{\partial}{\partial\theta}M}{24\,ACM} + \frac{\frac{\partial^2}{(\partial\theta)^2}M}{6\,CM} + \frac{\frac{\partial}{\partial\phi}A^2}{48\,A^2HM^2} - \frac{\frac{\partial^2}{(\partial\phi)^2}A}{24\,AHM^2}$$

$$\Psi_3 = \frac{\frac{\partial^2}{\partial r\partial\theta}A}{8\,A\sqrt{B}\sqrt{C}} - \frac{\frac{\partial}{\partial r}A\frac{\partial}{\partial\theta}A}{16\,A^2\sqrt{B}\sqrt{C}} - \frac{\frac{\partial}{\partial\theta}A\frac{\partial}{\partial t}B}{16\,A^{\frac{3}{2}}B\sqrt{C}} - \frac{\frac{\partial}{\partial\theta}A\frac{\partial}{\partial r}C}{16\,A\sqrt{B}C^{\frac{3}{2}}} + \frac{\frac{\partial}{\partial\theta}A\frac{\partial}{\partial t}H}{16\,A^{\frac{3}{2}}\sqrt{C}H} + \frac{\frac{\partial}{\partial t}C\frac{\partial}{\partial\theta}M}{8\sqrt{A}C^{\frac{3}{2}}M} + \frac{\frac{\partial}{\partial r}C\frac{\partial}{\partial\theta}M}{8\sqrt{B}C^{\frac{3}{2}}M} - \frac{\frac{\partial}{\partial t}H\frac{\partial}{\partial\theta}M}{8\sqrt{A}\sqrt{C}HM}$$
$$- \frac{\frac{\partial}{\partial r}H\frac{\partial}{\partial\theta}M}{8\sqrt{B}\sqrt{C}HM} - \frac{i\,\frac{\partial^2}{\partial r\partial\phi}A}{8\,A\sqrt{B}\sqrt{H}M} + \frac{i\,\frac{\partial}{\partial r}A\frac{\partial}{\partial\phi}A}{16\,A^2\sqrt{B}\sqrt{H}M} + \frac{i\,\frac{\partial}{\partial\phi}A\frac{\partial}{\partial t}B}{16\,A^{\frac{3}{2}}B\sqrt{H}M} - \frac{i\,\frac{\partial}{\partial\phi}A\frac{\partial}{\partial t}C}{16\,A^{\frac{3}{2}}C\sqrt{H}M} + \frac{i\,\frac{\partial}{\partial\phi}A\frac{\partial}{\partial r}H}{16\,A\sqrt{B}H^{\frac{3}{2}}M}$$

$$\Psi_4 = -\frac{\frac{\partial}{\partial\theta}A^2}{16\,A^2C} + \frac{\frac{\partial^2}{(\partial\theta)^2}A}{8\,AC} + \frac{\frac{\partial}{\partial t}A\frac{\partial}{\partial t}C}{16\,A^2C} + \frac{\frac{\partial}{\partial r}A\frac{\partial}{\partial t}C}{8\,A^{\frac{3}{2}}\sqrt{B}C} + \frac{\frac{\partial}{\partial t}B\frac{\partial}{\partial t}C}{16\,ABC} + \frac{\frac{\partial}{\partial t}C^2}{16\,AC^2} - \frac{\frac{\partial^2}{(\partial t)^2}C}{8\,AC} - \frac{\frac{\partial^2}{\partial t\partial r}C}{4\sqrt{A}\sqrt{B}C} + \frac{\frac{\partial}{\partial r}A\frac{\partial}{\partial r}C}{16\,ABC} + \frac{\frac{\partial}{\partial t}B\frac{\partial}{\partial r}C}{8\sqrt{A}B^{\frac{3}{2}}C} + \frac{\frac{\partial}{\partial r}B\frac{\partial}{\partial r}C}{16\,B^2C}$$
$$+ \frac{\frac{\partial}{\partial t}C\frac{\partial}{\partial r}C}{8\sqrt{A}\sqrt{B}C^2} + \frac{\frac{\partial}{\partial r}C^2}{16\,BC^2} - \frac{\frac{\partial^2}{(\partial r)^2}C}{8\,BC} - \frac{\frac{\partial}{\partial t}A\frac{\partial}{\partial t}H}{16\,A^2H} - \frac{\frac{\partial}{\partial r}A\frac{\partial}{\partial t}H}{8\,A^{\frac{3}{2}}\sqrt{B}H} - \frac{\frac{\partial}{\partial t}B\frac{\partial}{\partial t}H}{16\,ABH} - \frac{\frac{\partial}{\partial t}H^2}{16\,AH^2} + \frac{\frac{\partial^2}{(\partial t)^2}H}{8\,AH} + \frac{\frac{\partial^2}{\partial t\partial r}H}{4\sqrt{A}\sqrt{B}H} - \frac{\frac{\partial}{\partial r}A\frac{\partial}{\partial r}H}{16\,ABH} - \frac{\frac{\partial}{\partial t}B\frac{\partial}{\partial r}H}{8\sqrt{A}B^{\frac{3}{2}}H}$$
$$- \frac{\frac{\partial}{\partial r}B\frac{\partial}{\partial r}H}{16\,B^2H} - \frac{\frac{\partial}{\partial t}H\frac{\partial}{\partial r}H}{8\sqrt{A}\sqrt{B}H^2} - \frac{\frac{\partial}{\partial r}H^2}{16\,BH^2} + \frac{\frac{\partial^2}{(\partial r)^2}H}{8\,BH} - \frac{\frac{\partial}{\partial\theta}A\frac{\partial}{\partial\theta}M}{8\,ACM} - \frac{i\,\frac{\partial^2}{\partial\theta\partial\phi}A}{4\,A\sqrt{C}\sqrt{H}M} + \frac{i\,\frac{\partial}{\partial\theta}A\frac{\partial}{\partial\phi}A}{8\,A^2\sqrt{C}\sqrt{H}M} + \frac{i\,\frac{\partial}{\partial\phi}A\frac{\partial}{\partial\theta}M}{4\,A\sqrt{C}\sqrt{H}M^2} + \frac{\frac{\partial}{\partial\phi}A^2}{16\,A^2HM^2} - \frac{\frac{\partial^2}{(\partial\phi)^2}A}{8\,AHM^2}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "I"